Overpartition congruences modulo 17, 19 and 23: proofs of three conjectures of Wei

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Abstract

Wei recently proved Ramanujan-type congruences for the overpartition function $\overline{p}(n)$ modulo 11 and 13, and stated three further congruences. Modulo 17 and 23 they were expected to follow from his method after Sturm-bound computations involving more than 10^9 coefficients, which were not completed. Modulo 19 the congruence was stated to be beyond the methods of the paper. We prove all three. For the moduli 17 and 23 we compute the Hecke operator $T(\ell^2)$ on a space of dimension at most 6 and isolate a correction term that vanishes on the relevant residue class; the computation needs fewer than 4000 coefficients. For the modulus 19 the relevant generating function is only weakly holomorphic. We multiply it by two eta-quotients congruent to 1 modulo 19 before projecting onto the required classes, and show that the projection is a cusp form of weight $35/2$ on $\Gamma_0(23104)$. The congruence then follows from a $T(17^2)$ eigen-relation checked up to the Sturm bound.

1 Introduction

Let $\overline{p}(n)$ denote the number of overpartitions of n [2], so that $\sum_{n\geq 0}\overline{p}(n)q^n = 1/\varphi(-q)$ with $\varphi(q)=\sum_{k\in\mathbb{Z}} q^{k^2}$. Wei [8] proved infinite families of congruences for $\overline{p}$ modulo 11 and 13 by Hecke operators on the spaces $M_k(\Gamma_0(4))$, which are spanned by products of φ and $F(q)=\sum_{n \text{ odd}}\sigma(n)q^n$. In his closing section he recorded the following.

Theorem 1. *For all $n\geq 0$,*

$$\overline{p}\big(17\cdot 11^2(88n+m)\big)\equiv 0 \pmod{17}, \qquad m\in\{19,35,43,51,83\},$$
$$\overline{p}\big(23\cdot 13^2(104n+l)\big)\equiv 0 \pmod{23}, \qquad l\in\{29,53,61,69,77,101\}.$$

Theorem 2. $\overline{p}(17^2\cdot 4n)\equiv 0 \pmod{19}$ *for every integer $n\geq 0$ with $n\equiv 3 \pmod 8$, $\left(\frac{n}{17}\right)=-1$ and $\left(\frac{n}{19}\right)=1$.*

Wei states the congruences of Theorem 1, explains that they reduce to Sturm-bound checks of about $1.2\cdot 10^9$ and $4.0\cdot 10^9$ coefficients on $\Gamma_1(340736)$ and $\Gamma_1(562432)$, and notes that these computations had not been completed. He states Theorem 2 as a conjecture that cannot be proved by the methods of [8]. A further conjecture of [8], modulo 7, is treated in the modulo-7 paper in this versioned collection.

Contribution. For Theorem 1, Wei had already reduced the statements to finite computations and expected them to follow from his method. Our contribution there is a short structural argument replacing a computation of more than 10^9 coefficients by one with fewer than 4000. Theorem 2 settles a conjecture that [8] places beyond its methods.

Relation to general results. Treneer's pole-removal method [7] is a standard input. Ryan, Sirolli, Villegas-Morales and Zheng [6, Theorems 1.3 and 1.5] already give square-prime, Legendre-restricted congruences as well as cube-prime families. A cube-versus-square distinction would therefore be incorrect. The comparison must use the actual parameters: our modulo-19 target arguments are prime to 19, whereas their displayed modulo-19 families contain that factor; the modulo-17/23 targets have the specified small primes and additional classes modulo 8. Wei already supplies their starting forms in his closing section. The remaining contribution is the stated proof and computational reduction, not the invention of those forms or of square-prime congruences. The consulted sources did not supply these exact conclusions; this is a bounded priority assessment.

The residue classes in Theorem 1 are exactly the N with $N \equiv 3 \pmod 8$ and $\left(\frac{N}{11}\right) = -1$, respectively $N \equiv 5 \pmod 8$ and $\left(\frac{N}{13}\right) = 1$. This was checked by enumeration modulo 88 and 104.

Method. Both proofs use Shimura's operators $T(\ell^2)$ on half-integral weight forms [4, 3]. For $f = \sum a(n)q^n \in M_{\lambda+1/2}(\Gamma_0(N))$ (trivial character, $4 \mid N$, $\ell \nmid N$),

$$(f \mid T(\ell^2))(n) = a(\ell^2 n) + \left(\frac{(-1)^\lambda n}{\ell}\right) \ell^{\lambda-1} a(n) + \ell^{2\lambda-1} a(n/\ell^2), \tag{1}$$

with $a(x) = 0$ for $x \notin \mathbb{Z}$. Theorem 1 needs only the low-dimensional spaces $M_{\lambda+1/2}(\Gamma_0(4))$. Theorem 2 concerns $\overline{p}(4n)$, whose generating function $\varphi(q)^3/\varphi(-q)^4$ has a pole at the cusp 0. The new ingredient is a holomorphic representative modulo 19 of the relevant projection.

Sturm's theorem. If $g \in M_k(\Gamma_0(N), \chi)$ has p-integral coefficients and $a(n) \equiv 0 \pmod p$ for $n \le \frac{k}{12}[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(N)]$, then $g \equiv 0 \pmod p$ [5]. For half-integral weight we apply it to g^2 (or $g\varphi$), which has integral weight. If the first nonzero coefficient of g modulo p is at n_0, then that of g^2 is at $2n_0$, because $\mathbb{F}_p[[q]]$ is an integral domain.

2 Proof of Theorem 1

Write $(p, \ell, \lambda, r) = (17, 11, 7, 3)$ or $(23, 13, 10, 5)$.

Lemma 3 (Wei's reduction)**.** *There is $c = \sum c(n)q^n \in M_{\lambda+1/2}(\Gamma_0(4))$ with integer coefficients and $\sum \overline{p}(pn)(-q)^n \equiv c \pmod p$. Explicitly, with F as above,*

$$c = 13F^2\varphi^7 + 13F\varphi^{11} + \varphi^{15} \qquad (p = 17),$$
$$c = 20F^5\varphi + 17F^4\varphi^5 + 14F^3\varphi^9 + 5F^2\varphi^{13} + 9F\varphi^{17} + \varphi^{21} \qquad (p = 23).$$

Proof. We have $\sum \overline{p}(n)(-q)^n = 1/\varphi(q)$ and $\varphi(q)^p \equiv \varphi(q^p) \pmod p$, so $\varphi(q^p)\sum \overline{p}(n)(-q)^n \equiv \varphi^{p-1}$. Extracting the exponents divisible by p and replacing q^p by q gives $\varphi(q)\sum \overline{p}(pn)(-q)^n \equiv \varphi^{p-1} \mid T(p) \pmod p$, where $T(p)$ is the Hecke operator on $M_{(p-1)/2}(\Gamma_0(4), \chi_{-4}^{(p-1)/2})$ (its second term carries $p^{(p-3)/2} \equiv 0$). We computed $\varphi^{p-1} \mid T(p)$ exactly in the triangular basis $F^a\varphi^{p-1-4a}$. Its coordinates modulo p are $(1, 13, 13, 0, 0)$ for $p = 17$ and $(1, 9, 5, 14, 17, 20)$ for $p = 23$. Each is φ times the stated c. For $p = 17$ the coordinate of F^4 is 0, so no φ^{-1} term arises. Since $\varphi \in 1 + 2q\mathbb{Z}[[q]]$ is a unit modulo p, the claim follows. □

Proposition 4. *Modulo p, $c \mid T(\ell^2) \equiv \ell^{\lambda-1}c + h$ with $\ell^{\lambda-1} \equiv 8 \pmod{17}$, respectively $3 \pmod{23}$, and*

$$h = 12(F\varphi^{11} + F^2\varphi^7) \qquad (p = 17),$$
$$h = 16F\varphi^{17} + 16F^2\varphi^{13} + 14F^3\varphi^9 + 7F^4\varphi^5 + 15F^5\varphi \qquad (p = 23).$$

Proof. $T(\ell^2)$ preserves $M_{\lambda+1/2}(\Gamma_0(4))$, which has the upper-triangular basis $F^a\varphi^{2\lambda+1-4a}$ of dimension 4, respectively 6. We computed the integer matrix of $T(\ell^2)$ in this basis from (1), checking membership of each image in the span on all available coefficients. We then reduced $c \mid T(\ell^2) - \ell^{\lambda-1}c$ modulo p. □

Lemma 5. *$h(n) \equiv 0 \pmod p$ for all $n \equiv r \pmod 8$.*

Proof. Let $g = \sum_{n\equiv r\ (8)} h(n)q^n$. Since r is odd, averaging the twists of h by the four real characters modulo 8 shows $g \in M_{\lambda+1/2}(\Gamma_0(256))$ [8, Prop. 2.4 and Remark 2.5]. Then $g\varphi \in M_{\lambda+1}(\Gamma_0(256), \chi_{-4}^{\lambda+1})$ has integer coefficients. The Sturm bound is $(\lambda+1)\cdot 384/12 = 256$, respectively 352. The coefficients of $g\varphi$ vanish modulo p for $n < 4000$, so $g\varphi \equiv 0$, and $g \equiv 0$ because φ is a unit. □

Proof of Theorem 1. Let $N \equiv r \pmod 8$ with $\left(\frac{N}{\ell}\right) = -1$ (for $p = 17$), respectively $+1$ (for $p = 23$). Then $\ell \nmid N$, and the character value in (1) is $\left(\frac{-1}{11}\right)\left(\frac{N}{11}\right) = (-1)(-1) = 1$, respectively $\left(\frac{N}{13}\right) = 1$ since $\lambda = 10$ is even. Comparing N-th coefficients in Proposition 4,

$$c(\ell^2 N) + \ell^{\lambda-1}c(N) \equiv \ell^{\lambda-1}c(N) + h(N) \pmod p,$$

so $c(\ell^2 N) \equiv h(N) \equiv 0$ by Lemma 5. By Lemma 3, $\overline{p}(p\ell^2 N) \equiv (-1)^{\ell^2 N}c(\ell^2 N) \equiv 0 \pmod p$. □

3 Proof of Theorem 2

Let $c(n) = \overline{p}(4n)$. A 2-dissection gives

$$F_4 := \sum_{n\geq 0} c(n)q^n = \frac{\varphi(q)^3}{\varphi(-q)^4} = \frac{\eta(2z)^{19}}{\eta(z)^{14}\eta(4z)^6},$$

The identity follows algebraically, not from a finite check: with $E(\sum a_n q^n) = \sum a_{2n}q^n$, the classical relations $\varphi(q)\varphi(-q) = \varphi(-q^2)^2$, $E\varphi = \varphi(q^2)$ and $E(\varphi^2) = \varphi^2$ give successively

$$E(1/\varphi(-q)) = \varphi(q^2)/\varphi(-q)^2, \qquad E(\varphi(q^2)\varphi(q)^2/\varphi(-q^2)^4) = F_4.$$

Since $\varphi(q)$ and $\varphi(-q) = 2\varphi(q^4) - \varphi(q)$ transform under $\Gamma_0(16)$ with Shimura's theta multiplier $j(\gamma, z)$, F_4 is weakly holomorphic of weight $-1/2$. Its raw exponent at odd-denominator cusps is $-1/4$ (multiply by cusp width for the local-parameter order). Theorem 2 asserts $c(289n) \equiv 0 \pmod{19}$ on the stated classes.

A representative without poles at most cusps. Put $E_1 = \eta(z)^{19}/\eta(19z)$ and $E_2 = \eta(19z)^{19}/\eta(361z)$. Both lie in $1 + 19q\mathbb{Z}[[q]]$, because $(1-q^n)^{19} \equiv 1 - q^{19n}$. By the criterion of Gordon–Hughes–Newman and Ligozat, $E_1E_2 \in M_{18}(\Gamma_0(361))$ with trivial character. Set

$$F'' = F_4E_1E_2 = \frac{\eta(z)^5\eta(2z)^{19}\eta(19z)^{18}}{\eta(4z)^6\eta(361z)} \equiv F_4 \pmod{19}.$$

Then $F''(\gamma z) = j(\gamma, z)^{35}F''(z)$ on $\Gamma_0(16\cdot 361)$. Indeed $F_4(\gamma z) = j(\gamma, z)^{-1}F_4(z)$, and $(E_1E_2)(\gamma z) = (cz+d)^{18}(E_1E_2)(z)$. Since $j(\gamma, z)^2 = \left(\frac{-1}{d}\right)(cz+d)$, we have $(cz+d)^{18} = j(\gamma, z)^{36}$. So F'' is weakly holomorphic of weight $35/2$ with trivial character. Multiplying by an eta-quotient congruent to 1 modulo p to remove poles is the standard device of Treneer [7]. The point here is that the multiplication must precede the projection. At a cusp a/c with $361 \nmid c$, the factor E_1 (if $19 \nmid c$) or E_2 (if $19 \parallel c$) has order large enough to cancel the pole of F_4. So F'' has positive order at every cusp with $361 \nmid c$.

Projection. Let $G'' = \sum c''(n)q^n$, summed over $n \equiv 3 \pmod 8$ with $\left(\frac{n}{19}\right) = 1$, where $c''(n)$ are the coefficients of F''. Then $G'' \in \mathbb{Z}[[q]]$ and $G'' \equiv \sum_{n\equiv 3\,(8),\,(n/19)=1} c(n)q^n \pmod{19}$. Explicitly $G''(z) = \sum_{t \bmod 152} W_t F''(z + t/152)$. Writing $t = 19j + 8k$, the weights are $W_t = \frac{1}{8}e(-3j/8)\,b_k$ with $b_0 = 9/19$ and $b_k = -(1 + \chi(k)g)/38$, where $\chi = \left(\frac{\cdot}{19}\right)$ and g is the Gauss sum. These weights reproduce the indicator function exactly.

Lemma 6. *$G''(\gamma z) = j(\gamma, z)^{35} G''(z)$ for all $\gamma \in \Gamma_0(23104)$, where $23104 = 64 \cdot 361$.*

Proof. Let $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(23104)$, so $152^2 \mid c$. For each t put $t' \equiv td^2 \pmod{152}$. Then $\gamma' = T_{t/152}\gamma T_{-t'/152}$ is in $\Gamma_0(16 \cdot 361)$ with lower-right entry $d' = d - ct'/152 \equiv d$ modulo 8 and modulo $c/23104$. Hence $j(\gamma', z + t'/152) = j(\gamma, z)$. Since $d^2 \equiv 1 \pmod 8$ and $\chi(kd^2) = \chi(k)$, we have $W_{t'} = W_t$, and the sum is permuted. □

Proposition 7. *G'' is a cusp form: $G'' \in S_{35/2}(\Gamma_0(23104))$.*

Proof. At the 228 cusps with $361 \nmid c$, every translate is expanded at a cusp whose reduced denominator is again not divisible by 361; exact order arithmetic gives strictly positive order in all $228 \cdot 152 = 34656$ cases. At the other twelve cusps, some translated summands have nonpositive exponents; this does not say that F'' itself has a pole at each of them. Appendix A specifies the complete finite obligation and expansion convention. Grouping by exponent and rational square class leaves 28 cyclotomic sums with 32928 summands. Every sum reduces exactly to zero modulo its cyclotomic polynomial. Thus all principal parts and constant terms vanish. This exact calculation, not numerical decay, proves the assertion.

Historical numerical corroboration used a separately written program; it is not the exact proof or a newly repeated scan. It evaluated $|G''(\gamma z)(cz + d)^{-35/2}|$ at 200-digit precision at all 240 cusps, along vertical rays of increasing height. It observed exponential decay everywhere, with stable fitted orders of 16, 19, 4 and 11 in the local parameter, and found no constant term or pole. At the 12 critical cusps the cancellation is by factors 10^{-63} to 10^{-118}, well within the working precision. The same program checked the transformation law on several matrices of $\Gamma_0(23104)$. □

Proof of Theorem 2. Since $17 \nmid 23104$, $T(289)$ acts on $S_{35/2}(\Gamma_0(23104))$ by (1) with $\lambda = 17$. Moreover $17^{16} \equiv 5$ and $17^{33} \equiv 17^{15} \pmod{19}$. Let $H = G'' \mid T(289) - 14G''$, a cusp form with integer coefficients. Its square belongs to $S_{35}(\Gamma_0(23104), \chi_{-4})$. The index is 36480, so the character-version Sturm bound is 106400. It suffices to check $H(n) \equiv 0 \pmod{19}$ for $0 \le n \le 53200$, inclusively. One implementation generates F_4 through exponent $289 \cdot 53200 = 15374800$ (with ten spare coefficients); a separate implementation generates $\overline{p}$ through argument $4 \cdot 289 \cdot 53200 = 61499200$. Both test the same finite obligation, with no exception. Hence, for all n in the support,

$$c(289n) + 5\left(\frac{n}{17}\right)c(n) + 17^{15}c(n/289) \equiv 14\,c(n) \pmod{19}.$$

If $n \equiv 3 \pmod 8$, $\left(\frac{n}{19}\right) = 1$ and $\left(\frac{n}{17}\right) = -1$, then $17 \nmid n$, $289n$ lies in the same classes, and the relation gives $c(289n) \equiv 19c(n) \equiv 0$. □

Remark 8. The scalar 14 differs from $1 + 17^{33} \equiv 8 \pmod{19}$; no particular Eisenstein series is being identified here. The relation does not extend to arbitrary coefficients outside the projection. Projecting F_4 before multiplying by E_1E_2 leaves poles, so the order of the two operations matters.

4 Computations

All computations use exact integer or modular arithmetic (python-flint), except the high-precision confirmation in Proposition 7. The scripts and logs are in the accompanying evidence bundle. Each theorem was also confirmed directly: Theorem 1 for arguments up to $2 \cdot 10^7$ (552 and 295 cases), and Theorem 2 for all $n \le 53200$.

A Exact cusp certificate

Write $e(x) = \exp(2\pi i x)$. The eta-exponent vector, in increasing argument order, is

$$(\delta, r_\delta) = (1,5), (2,19), (4,-6), (19,18), (361,-1).$$

Represent the cusps by a/c with $c \mid L = 23104$, $\gcd(a,c) = 1$, and one numerator in each unit class modulo $\gcd(c, L/c)$. There are 240 such cusps. Choose $A_s = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ with $ad - bc = 1$. The expansion convention is

$$(-i(c\tau + d))^{-35/2} G''(A_s\tau) = \sum_E C_E e(E\tau).$$

The cusp width is $w = L/\gcd(c^2, L)$, so the exponent in the width-normalised local parameter is wE. Multiplication by a nonzero phase to change slash conventions does not change vanishing.

c	cusps	w	raw $E \le 0$	groups
361	1	64	$-1/4, 0$	3
722	1	16	$-1, 0$	3
1444	2	4	$-4, 0$	6
2888	4	1	$-16, 0$	12
5776	2	1	0	2
11552	1	1	0	1
23104	1	1	0	1
Total	12			28

The listed exponents are those arising from translated summands before cancellation. All resulting group sums are zero.

Matrix and multiplier convention. For each $t = 19j + 8k$, decompose the integer matrix

$$M_t = \begin{pmatrix} 152a + tc & 152b + td \\ 152c & 152d \end{pmatrix} = \gamma \begin{pmatrix} u_1 & u_2 \\ 0 & u_3 \end{pmatrix}, \qquad \gamma \in \mathrm{SL}_2(\mathbb{Z}),\ u_1, u_3 > 0.$$

This is obtained by the extended Euclidean algorithm on the first column, with u_1 its positive gcd. For each eta factor similarly decompose $\mathrm{diag}(\delta, 1)\gamma = \gamma_\delta V_\delta$. For a matrix $(a', b'; c', d')$ with $c' > 0$, use

$$\eta(\gamma_\delta z) = e\left(\frac{a' + d'}{24c'} - \frac{s(d', c')}{2}\right)[-i(c'z + d')]^{1/2}\eta(z),$$

where s is the rational Dedekind sum and the analytic square root is fixed on the upper half-plane. This is the convention in Kong–Teo [1, Eq. (1.2)] after absorbing the $e(-1/8)$ phase into the square root.

Finite truncation and exact arithmetic. The decompositions give a leading exponent E_0, a root-of-unity phase, a positive square root of a rational number, and factors

$$\prod_{\delta}\prod_{n\geq 1}(1-e(n\beta_\delta)q^{n\alpha_\delta})^{r_\delta}, \qquad \alpha_\delta > 0.$$

If $E_0 > 0$, no term is needed. Otherwise expand only increments at most $-E_0$. All increments are positive; consequently omitted factors and higher binomial terms cannot contribute to an exponent $E \leq 0$. For $r < 0$ use $\binom{r}{m} = (-1)^m\binom{-r+m-1}{m}$. All phases and amplitudes before square roots are rational and retained exactly.

Multiply by the specified projection weights W_t and group by E and rational square class. Within each group, choose the least common denominator n of phases and form $P(X) = \sum_j q_j X^{nr_j \bmod n} \in \mathbb{Q}[X]$. Check $P \bmod \Phi_n = 0$. The actual fields have $n = 38, 76, 152$. This separate vanishing of all square-class groups is sufficient; no assumption of their linear independence is required. The certificate contains 32928 summands in 28 groups. Projection denominators divisible by 19 cause no reduction problem: G'' is defined as retention of integer coefficients, and the Fourier weights are used only to prove its modularity and cusp behaviour in characteristic zero.

Reproduction. From the collection root run `python verify.py mod17-19-23`. The wrapper pins the lifted eta vector, checks all 34656 positive orders, requires exactly twelve critical cusps and all 28 zero groups, and performs the inclusive Hecke–Sturm check. It fails on an unmet obligation, not merely prints a Boolean. Numerical cusp decay is retained as corroboration only.

Proposition 9 (Sufficient lift–project criterion)**.** *Let f be weakly holomorphic with integral Fourier coefficients, and let $E \equiv 1 \pmod{p}$ be an integral eta product or quotient. Suppose a finite Fourier projection P retains specified integer coefficients, $P(Ef)$ has a proved modular transformation law at a finite-index level, and every nonpositive cusp coefficient vanishes by complete finite expansions as above. Then $P(Ef)$ is an integral cusp form congruent to $P(f)$ modulo p. Any proposed Hecke congruence for it follows from a valid Sturm check in the resulting weight, level and character.*

Proof. The transformation law and cusp expansions establish cusp membership. Coefficient retention commutes with reduction, so $P(Ef) \equiv P(f)$. Apply the relevant Hecke operator and Sturm theorem to their difference. □

This is a sufficient certificate pattern, not an automatic existence theorem for a suitable E, nor a claim that this general principle is new.

Collection and verification

This paper is part of the revised collection, version 1.1.0-candidate, DOI `https://doi.org/10.5281/zenodo.23319157`. Companion papers and exact verification code are archived together. The statements are unrefereed computer-assisted proof candidates; no formal verification or unaffiliated reproduction is asserted.

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